API Latency Report

Python

BASE_URL=http://localhost:8000 | K_RPS=50 | RESTORE_N=1000000 | DURATION=10s

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /ping	500	1.57	21.03	5.68	5.56	5.56	8.08	9.4	12.9	19.23
GET /user/{id}/score	500	2.19	115.75	7.92	6.57	6.57	9.87	12.08	57.94	114.83
GET /user/{id}/rank	500	1.77	419.27	21.58	6.87	6.87	12.23	95.87	399.66	418.04
GET /score/{score}/count	500	2.39	310.65	13.47	5.86	5.86	10.25	39.02	221.06	291.27
GET /leaderboard	500	4.23	198.76	12.19	8.93	8.93	13.34	16.8	127.68	183.96
GET /leaderboard?min_rank=&max_rank= (dynamic)	17	3543.0	29306.16	14816.21	14382.94	14382.94	26593.92	27343.28	28913.58	29266.9
GET /leaderboard?min_score=&max_score= (dynamic)	no successful samples									
GET /leaderboard/all	no successful samples									

Golang

BASE_URL=http://localhost:8080 | K_RPS=50 | RESTORE_N=1000000 | DURATION=10s

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /ping	500	1.30	106.12	3.10	2.05	2.05	3.77	5.25	13.41	97.43
GET /user/{id}/score	500	1.34	191.93	4.43	2.07	2.07	3.52	5.26	92.20	182.11
GET /user/{id}/rank	500	1.48	72.15	3.58	2.22	2.22	5.14	7.64	35.90	64.15
GET /score/{score}/count	500	1.44	49.40	3.08	2.27	2.27	3.84	5.23	19.00	46.71
GET /leaderboard	500	1.55	83.17	3.12	2.36	2.36	3.94	4.90	11.65	72.89
GET /leaderboard?min_rank=&max_rank= (dynamic)	30	1087.39	29707.10	10520.74	9623.74	9623.74	26849.49	29060.32	29597.48	29696.14
GET /leaderboard?min_score=&max_score= (dynamic)	18	1532.11	29624.37	15783.87	16667.88	16667.88	21913.50	23079.69	28315.43	29493.47
GET /leaderboard/all	no successful samples									

In-Memory (Fenwick)

BASE_URL=http://localhost:8081 | K_RPS=50 | RESTORE_N=1000000 | DURATION=10s

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /ping	500	1.49	24.76	2.95	2.54	2.54	4.35	5.78	9.68	19.16
GET /user/{id}/score	500	1.32	18.83	4.69	4.37	4.37	6.75	8.56	12.25	17.63
GET /user/{id}/rank	500	1.41	39.58	4.92	4.40	4.40	7.25	9.41	15.81	39.13
GET /score/{score}/count	500	1.59	61.45	5.52	4.36	4.36	7.95	11.83	32.97	59.34

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /leaderboard	500	1.57	33.27	5.27	4.86	4.86	8.11	9.67	15.46	31.46
GET /leaderboard?min_rank=&max_rank= (dynamic)	29	274.21	10684.67	3945.54	3577.09	3577.09	8074.85	8416.61	10111.87	10627.39
GET /leaderboard?min_score=&max_score= (dynamic)	470	1.23	1005.67	12.73	4.05	4.05	6.39	7.90	16.27	1005.21
GET /leaderboard/all	500	1.45	55.37	5.24	4.76	4.76	7.18	8.78	16.90	45.85

Direct (Fenwick)

Direct calls (no HTTP) | K_RPS=50 | RESTORE_N=1000000 | DURATION=10s

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /ping	500	0.00	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.03
GET /user/{id}/score	500	0.00	4.12	0.08	0.03	0.03	0.12	0.17	0.96	3.74
GET /user/{id}/rank	500	0.00	8.41	0.12	0.04	0.04	0.12	0.24	1.90	6.94
GET /score/{score}/count	500	0.00	0.70	0.00	0.00	0.00	0.00	0.00	0.06	0.40
GET /leaderboard	500	0.00	7.34	0.04	0.01	0.01	0.06	0.09	0.18	4.10
GET /leaderboard?min_rank=&max_rank= (dynamic)	500	0.42	54.32	5.37	3.88	3.88	8.59	17.35	28.19	49.25
GET /leaderboard?min_score=&max_score= (dynamic)	500	2.34	64.79	6.71	4.32	4.32	13.83	15.69	19.70	51.26

Endpoint	Count	Min (ms)	Max (ms)	Avg (ms)	Median (ms)	p50 (ms)	p90 (ms)	p95 (ms)	p99 (ms)	p99.9 (ms)
GET /leaderboard/all	500	2.56	49.67	10.08	4.69	4.69	25.69	32.53	46.69	48.75

Avg (ms) by endpoint – comparison

Performance comparison (% improvement)

Positive % = faster than baseline. Baseline: Python (Redis).

Endpoint	Python (Redis) ms	Go (Redis) ms	Go (Fenwick) ms	Direct (Fenwick) ms	Go-Redis vs Python	Go-Fenwick vs Python	Direct vs Python	Direct vs Go-Fenwick
GET /ping	5.68	3.10	2.95	0.00	45.4%	48.1%	100.0%	100.0%
GET /user/{id}/score	7.92	4.43	4.69	0.08	44.1%	40.8%	99.0%	98.3%
GET /user/{id}/rank	21.58	3.58	4.92	0.12	83.4%	77.2%	99.4%	97.6%
GET /score/{score}/count	13.47	3.08	5.52	0.00	77.1%	59.0%	100.0%	100.0%
GET /leaderboard	12.19	3.12	5.27	0.04	74.4%	56.8%	99.7%	99.2%
GET /leaderboard?min_rank=&max_rank= (dynamic)	14816.21	10520.74	3945.54	5.37	29.0%	73.4%	100.0%	99.9%
GET /leaderboard?min_score=&max_score= (dynamic)	-	15783.87	12.73	6.71	-	-	-	47.3%
GET /leaderboard/all	-	-	5.24	10.08	-	-	-	-92.4%